

Alexander Ticket

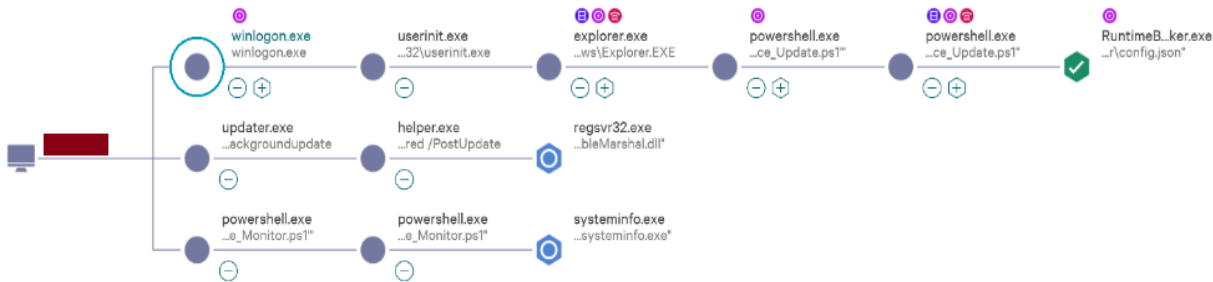
Note: This report has been sanitized for public sharing.

All internal IPs, hostnames, and CrowdStrike URLs have been redacted or replaced with simulated values.

Report was originally prepared for Jira; internal console links are not publicly accessible. Query references shown for context

The Summary :

The alert indicates a sequence of suspicious activities including defense evasion through process masquerading of a core Windows binary (**RuntimeBroker.exe**) executed from an anomalous user directory, detection of an untrusted payload via cloud-based machine learning, living-off-the-land proxy execution via **regsvr32.exe** beneath a browser process subtree, and local host reconnaissance using **systeminfo.exe** to gather infrastructure and OS vulnerabilities on host [redacted-host].



Confidence - Medium

Description

47/100

This automated lead detected the following behaviors:

- A file appears to be imitating a standard OS or otherwise benign filename and/or launched from an unusual location. This might be to masquerade malware. Review the file.
- A file meets the cloud-based machine learning low confidence threshold for malicious files. Detection is based on a high degree of entropy, packing, anti-malware evasion, or other similarity to known malware.
- Admin/recon tool in subtree beneath a browser process
- A process gathered information about the operating system or hardware. Adversaries can use this to identify system vulnerabilities. Review the process tree.

Show less

Techniques: Masquerading, Cloud-based ML, System Information Discovery

Involved hosts: FINANCE

Detections and context 4 items

Severity	Detect time	Process on host	Type	Command line	Tactic via technique
 --	2026/05/05 12:02:51	systeminfo.exe on [redacted] by [redacted]	Automated le...	"C:\Windows\system32\systeminfo.exe"	Discovery via System Information Disc...
 --	2026/05/04 14:59:34	RuntimeBroker.exe on [redacted] by [redacted]	Automated le...	"C:\Users\[redacted]\AppData\Local\Micr...	Machine Learning via Cloud-based ML
 --	2026/05/04 14:59:34	regsvr32.exe on [redacted] by [redacted]	Automated le...	"C:\Windows\system32\regsvr32.exe" /...	Discovery via System Information Disc...
 Medium	2026/05/04 07:16:52	RuntimeBroker.exe on [redacted] by [redacted]	Endpoint det...	"C:\Users\[redacted]\AppData\Local\Micr...	Defense Evasion via Masquerading

CrowdStrike Link:

- **Incident(CrowdScore):**

(internal link, redacted for confidentiality)

- **1. Endpoint Detection (Defense Evasion via Masquerading - RuntimeBroker.exe) :**

(internal link, redacted for confidentiality)

- **2. Automated Lead Context (Machine Learning via Cloud-based ML) :**

(internal link, redacted for confidentiality)

Description

A multi-stage intrusion was detected on host [redacted-host], characterized by an initial credential-based Remote Desktop (RDP) session hijacking followed by secondary payload execution via the local mail client. The adversary gained initial access at 06:27:19 by reconnecting to a long-standing, persistent **REMOTE_INTERACTIVE** session of the '[redacted]' user. Immediate defense evasion was conducted via **runonce.exe** to destroy installation artifacts, followed by the execution of an anomalous PowerShell stager disguised as **firefox_dump.zip** at 06:29 to harvest browser credentials.

At 07:13, operating within the hijacked desktop session, the adversary initiated an updated instance of Mozilla Thunderbird (**thunderbird.exe**) to force an encrypted IMAP synchronization at 07:14:06. This step was utilized to ingest a targeted spear-phishing mail

containing a malicious shortcut file (**Runtime_Service_Update.ps1.lnk**) into the local profile database. At 07:15, this shortcut was executed, triggering a secondary script that performed binary masquerading by compile-writing a packed cryptomining implant (**RuntimeBroker.exe**) into an anomalous user path. Threat progression was terminated by CrowdStrike Falcon, which blocked the masqueraded process and quarantined the core execution file. Approximately 16 hours later, the adversary re-engaged the host via a separate script to perform automated host reconnaissance using **systeminfo.exe**.

- ◆ Host Information:

Hostname: **[redacted-host]**

Operating System: Windows 10

External IP: [redacted]

Local IP Address: [redacted internal IP]

Host Type: Workstation

- ◆ User Information:

1. Initial Compromised Account:

Username: **[redacted]**

Local Admin: Yes

Login Type: REMOTE_INTERACTIVE

Login Time: Feb. 2, 2026 09:57:05

Login Domain: **[redacted-domain]**

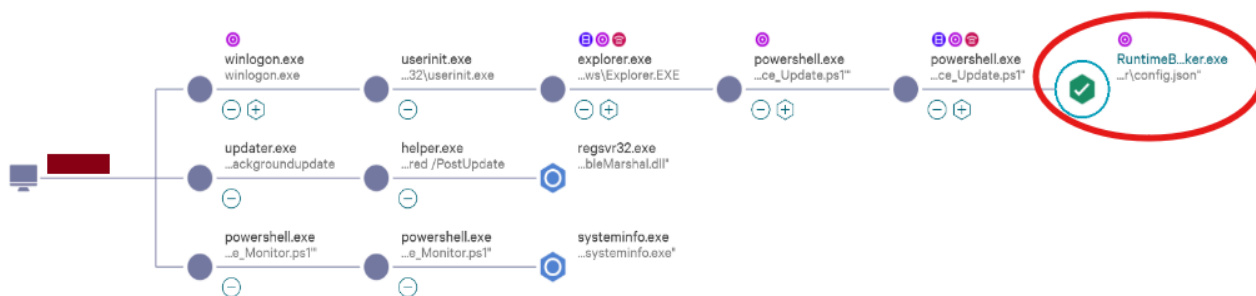
- ◆ Tactic & Technique:

Technique: Defense Evasion via Masquerading, Discovery via System Information
Discovery, Machine Learning via Cloud-based ML, Sensor-based ML

MITRE ATT&CK ID: **T1036, T1218.010, T1059.001, T1082**

Investigation Findings and Analysis:

Observed Command 1:



① Phase 1: Initial Access, Execution & Payload Deployment

Process: explorer.exe

Start time: May 4, 2026 06:27:19

Command line

C:\Windows\Explorer.EXE

File path

\Device\HarddiskVolume2\Windows\explorer.exe

Executable SHA256

988a56d897915315eef9ca679b3bc8adfcecf5e227aea99aaa1817620520e97e

[Resultthere]<https://www.virustotal.com/gui/file/988a56d897915315eef9ca679b3bc8adfcecf5e227aea99aaa1817620520e97e>]

0

/ 71

Community Score

File distributed by Microsoft

988a56d897915315eef9ca679b3bc8adfcecf5e227aea99aaa1817620520e97e

EXPLORER.EXE

peexe

64bits

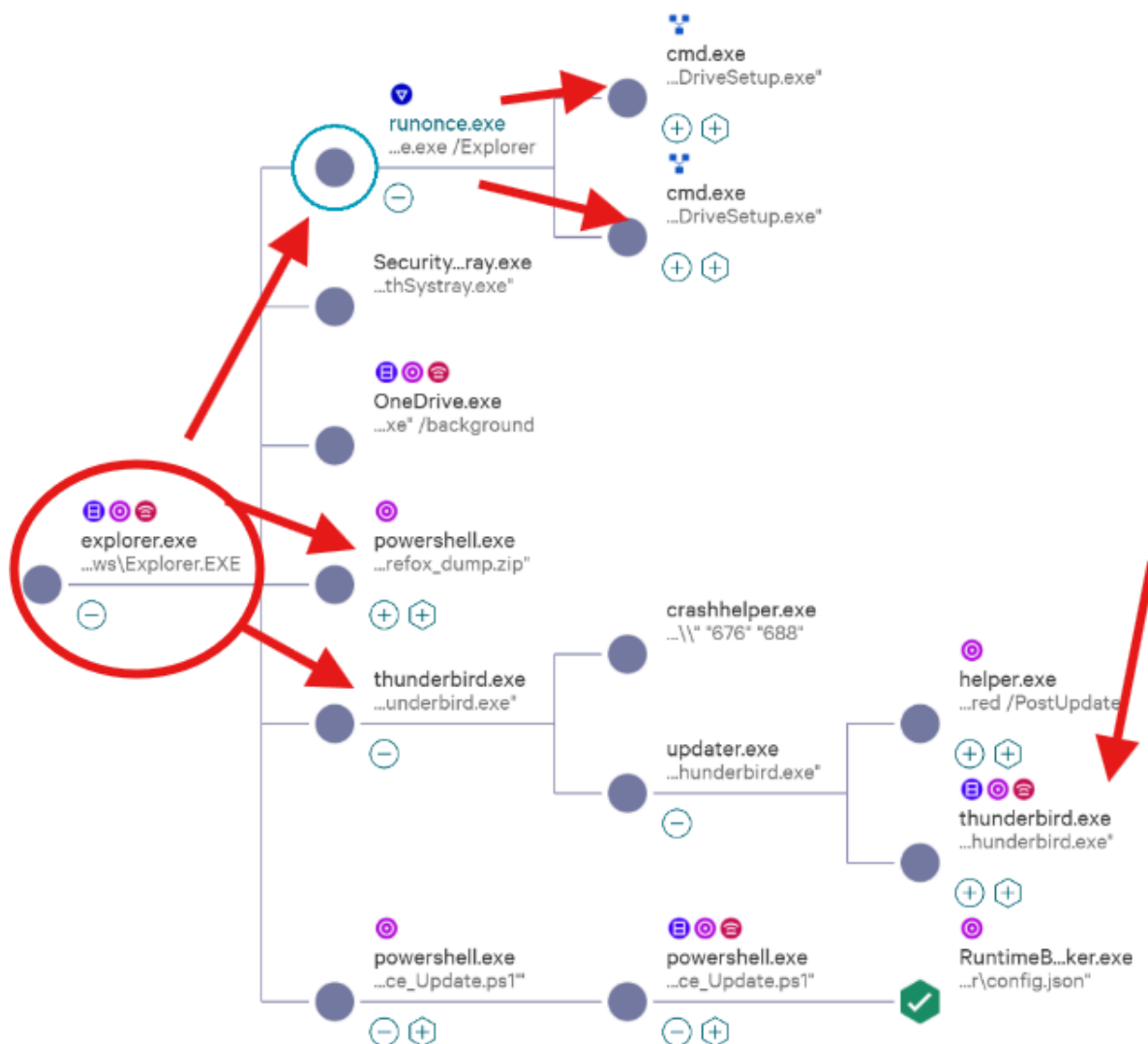
signed

known-distributor

overlay

Scanned	Detections	Type	Name
2025-11-14	0 / 66	ZIP	expl.zip
2026-05-25	9 / 60	RAR	test.rar
2026-03-23	0 / 66	ZIP	file_0357514d4e1843489e0452f7b044c3f0_2026-03-23_00_06_32_026000.zip
2026-05-20	13 / 69	Win32 EXE	m4bwdrz.exe
2026-04-01	48 / 71	Win32 EXE	testttt.exe
2025-11-06	0 / 61	RAR	VADS2.rar
2025-12-13	42 / 71	Win32 EXE	fud.exe
2025-12-06	31 / 72	Win32 EXE	SystemConfigurationUtility.exe
2025-12-11	45 / 71	Win32 EXE	crypted_app.exe
2026-04-01	48 / 71	Win32 EXE	crypted_app.exe
2026-01-21	0 / 62	RAR	saved_executables.rar
-	-	file	b469a86eb7bfe35bc41c1eb59fea76b8092dc67e43d9890d9f7fe25ca1451796
2025-12-10	42 / 71	Win32 EXE	crypted_app.exe
2025-12-08	41 / 72	Win32 EXE	perm dark.exe
2026-05-24	26 / 71	Win32 EXE	explorer
2025-12-18	0 / 62	ZIP	explorer.exe (1).zip
2026-01-07	38 / 71	Win32 EXE	sfss5l720.exe

The lineage begins with the primary user shell. This confirms the activity originated from an interactive RDP session. Although the binary is legitimate (0/71 on VT), the 11/17 malicious execution parents suggest this session context is frequently targeted for hijacking. The adversary utilized the existing [redacted-host] session to bypass the need for a new authentication event.



Subprocess: runonce.exe
Parent Process : explorer.exe

Start time: May 4, 2026 06:27:44

End time: May 4, 2026 06:27:51

Command line

"runonce.exe /Explorer"

File path

\Device\HarddiskVolume2\Windows\System32\runonce.exe

Executable SHA256

eb40639c488a46f8b787ab4ebc7a4f712286971432f1e8675e993dcc4c778554

[Resultthere|<https://www.virustotal.com/gui/file/eb40639c488a46f8b787ab4ebc7a4f712286971432f1e8675e993dcc4c778554/detection>]

Registry operations

ASEP Value Update

1

Time	Operation	Value Name
May 4, 2026 06:27:47	A key value was deleted	Delete Cached Update Binary
Key	\REGISTRY\MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\RunOnce	

This process represents a critical pivot point leveraging Windows Auto-Start Execution Points (ASEP) for defense evasion. Triggered directly by the user shell (**explorer.exe**), **runonce.exe** executed an entry staged in the registry under the **RunOnce** key titled **Delete Cached Update Binary**. The binary itself is a legitimate Windows component (0/69 on VT), but its execution was subverted by the adversary to sequentially spawn two command shell instances designed to purge forensic artifacts before the system's runtime monitoring could flag the source installer.

Subprocess: cmd.exe (Multiple Instances)
Parent Process : runonce.exe

Start time: May 4, 2026 06:27:46

End time: May 4, 2026 06:27:50

Command line 1

```
"C:\Windows\System32\cmd.exe" /q /c del /q "C:\Program Files\Microsoft OneDrive\Update\OneDriveSetup.exe"
```

Command line 2

```
"C:\Windows\System32\cmd.exe" /q /c del /q "C:\Program Files\Microsoft OneDrive\StandaloneUpdater\OneDriveSetup.exe"
```

File path

```
\Device\HarddiskVolume2\Windows\System32\cmd.exe
```

Executable SHA256

```
badf4752413cb0cbdc03fb95820ca167f0cdc63b597ccdb5ef43111180e088b0
```

[Result there|<https://www.virustotal.com/gui/file/badf4752413cb0cbdc03fb95820ca167f0cdc63b597ccdb5ef43111180e088b0/relations>]

Spawned as direct sub-processes of **runonce.exe**, these instances utilize the native Microsoft Command Shell binary, which displays a clean signature and high prevalence globally (0/53 on VT). However, CrowdStrike Falcon generated Medium and Low severity "Anomalous Process Execution" alerts targeting this specific sequence. The EDR's behavioral AI flagged the execution context due to highly suspicious command-line parameters optimized for stealth history suppression and quiet file deletion (**/q /c del**). This correlates with known multi-stage LNK exploitation tactics where trusted system binaries are abused to execute automated indicator removal, successfully purging the initial delivery payloads from the OneDrive directories to impede forensic timeline reconstruction.

Subprocess: powershell.exe
Parent Process : explorer.exe

Start time: May 4, 2026 06:29:08

End time: May 4, 2026 06:29:14

Command line

```
"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -w  
hidden -ep bypass -File  
"C:\Users\[redacted]\AppData\Local\Temp\firefox_dump.zip"
```

File path

```
\Device\HarddiskVolume2\Windows\System32\WindowsPowerShell\v1.0\powersh  
ell.exe
```

Executable SHA256

```
9785001b0dcf755eddb8af294a373c0b87b2498660f724e76c4d53f9c217c7  
a3
```

[Result there]<https://www.virustotal.com/gui/file/9785001b0dcf755eddb8af294a373c0b87b2498660f724e76c4d53f9c217c7a3/detection>]

Flagged by defensive systems due to highly evasive command-line arguments designed to conceal execution visibility from the user (**-w hidden**) and intentionally override corporate security configurations (**-ep bypass**). The adversary utilized a native, trusted Microsoft interpreter to execute code stored inside an obfuscated script masquerading as a compressed folder archive (**firefox_dump.zip**) within the local Temp directory. Community threat intelligence marks this execution context as highly suspicious, serving as the primary automated bridge to deploy the final masqueraded **RuntimeBroker.exe** implant.

Subprocess: thunderbird.exe (Updated Instance)
Parent Process : updater.exe

Start time: May 4, 2026 07:13:45

End time: May 4, 2026 07:15:57

Command line

```
"C:\Program Files\Mozilla Thunderbird\thunderbird.exe"
```

File path

```
\Device\HarddiskVolume2\Program Files\Mozilla  
Thunderbird\updated\thunderbird.exe
```

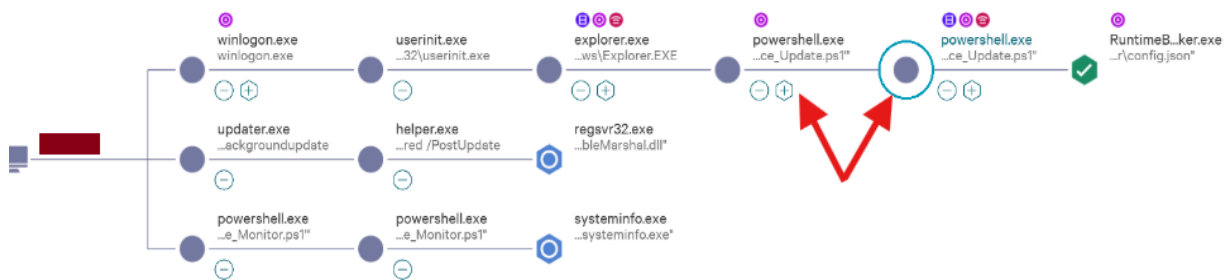
Executable SHA256

```
e414b612806f2b9b642e0979a75e4ee9de498b4ca8372b00c0deecc072e06f  
3f
```

[Result there]<https://www.virustotal.com/gui/file/e414b612806f2b9b642e0979a75e4ee9de498b4ca8372b00c0deecc072e06f3f>]

This specific process tracks an updated instance of the mail client running within the hijacked user session context. Timeline analysis confirms this process is not the initial network entry point, as malicious execution (**firefox_dump.zip**) occurred 44 minutes prior. Instead, this activity represents post-compromise deployment tradecraft. Network telemetry confirms that at 07:14:06, the adversary utilized the active session to force encrypted synchronization over port 993 (IMAP) with the internal mail server **[redacted]**. This maneuver allowed the attacker to

extract a pre-staged email containing the malicious shortcut payload (**Runtime_Service_Update.ps1.lnk**) directly into the local profile database, effectively using the compromised mailbox as an internal data staging and execution vector.



Process: powershell.exe (1)

Start time: May 4, 2026 07:15:38

End time: May 4, 2026 07:15:44

Command line

```
"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" "-Command"
"if((Get-ExecutionPolicy) -ne 'AllSigned') { Set-ExecutionPolicy
-Scope Process Bypass }; &
'C:\Users\[redacted]\Downloads\Runtime_Service_Update.ps1'"
```

File path

\Device\HarddiskVolume2\Windows\System32\WindowsPowerShell\v1.0\powershell.exe

Executable SHA256

9785001b0dcf755eddb8af294a373c0b87b2498660f724e76c4d53f9c217c7a3

[Resultthere]<https://www.virustotal.com/gui/file/9785001b0dcf755eddb8af294a373c0b87b2498660f724e76c4d53f9c217c7a3>]

0

/ 70

Community Score

-25

File distributed by Microsoft

9785001b0dcf755eddb8af294a373c0b87b2498660f724e76c4d53f9c217c7a3

PowerShell.EXE

peexe

long-sleeps

detect-debug-environment

known-distributor

64bits

legit

This process marks the initial automated triage phase of the script execution layout. Spawned directly from the user shell following the link file interaction, this instance evaluates the host's current script execution restrictions. By executing an inline command block, the script dynamically forces an execution policy override exclusively within its current process scope (**-Scope Process Bypass**), bypassing global administrative restrictions without altering permanent system registry values. This defense evasion maneuver was critical to immediately prime the system environment for running the un-signed **Runtime_Service_Update.ps1** script located inside the user's Downloads directory.

Process: powershell.exe (2)

Start time: May 4, 2026 07:15:43 End time: May 4, 2026 07:15:51

Command line
"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -NoProfile
-ExecutionPolicy Bypass -File
"C:\Users\[redacted]\Downloads\Runtime_Service_Update.ps1"

This process represents the core operational execution engine of the threat workflow, running with the highly evasive **-NoProfile** flag to block user-specific profile modifications from triggering security event logs. Network telemetry shows active automated external staging over port 443 (HTTPS) interacting directly with **github.com** and **release-assets.githubusercontent.com** (IPs: **140.82.121.4**, **185.199.110.133**) to stream dual-purpose payloads.

Disk and memory telemetry indicate a sophisticated multi-component drop:

- 1. Resource Hijacking: The process extracted and dropped a cryptocurrency mining suite (**xmrig.exe**, **solo_mine_example.cmd**) along with a low-level hardware access kernel-mode driver (**WinRing0x64.sys**) into the local user **Temp** path.
- 2. Persistence & Defense Evasion: Simultaneously, the script performed process masquerading by compile-writing an untrusted binary implant named **RuntimeBroker.exe** into an anomalous local application directory (**\Device\HarddiskVolume2\Users\[redacted]\AppData\Local\Microsoft\WindowsApps\RuntimeBroker\RuntimeBroker.exe**), laying the groundwork for an evasive execution tree mimicking standard OS background operations.

Remote	Port	Time
185.199.110.133	443	May 4, 2026 07:15:47

Remote	Port	Time
140.82.121.4	443	May 4, 2026 07:15:47

Filename	Time
<code>\Device\HarddiskVolume2\Users\[redacted]\AppData\Local\Temp\xmrig-6.21.0\solo_mine_example.cmd</code>	May 4, 2026 07:15:48

Filename	Time
\Device\HarddiskVolume2\Users\ [REDACTED] \AppData\Local\Temp\xmrig-6.21.0\xmrig.exe	May 4, 2026 07:15:48

Domain	Time
release-assets.githubusercontent.com	May 4, 2026 07:15:49
Domain	Time
github.com	May 4, 2026 07:15:48

Ip 140.82.121.4

[Resultthere|<https://www.virustotal.com/gui/ip-address/140.82.121.4>]

Community Score

1/91 security vendor flagged this IP address as malicious

140.82.121.4 (140.82.112.0/20)

AS 36459 (GitHub, Inc.)

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Security vendors' analysis ⓘ

Chong Lua Dao

Malicious

Ip 185.199.110.133

[Resultthere|<https://www.virustotal.com/gui/ip-address/185.199.110.133>]

Community Score

1/91 security vendor flagged this IP address as malicious

185.199.110.133 (185.199.108.0/22)

AS 54113 (Fastly, Inc.)

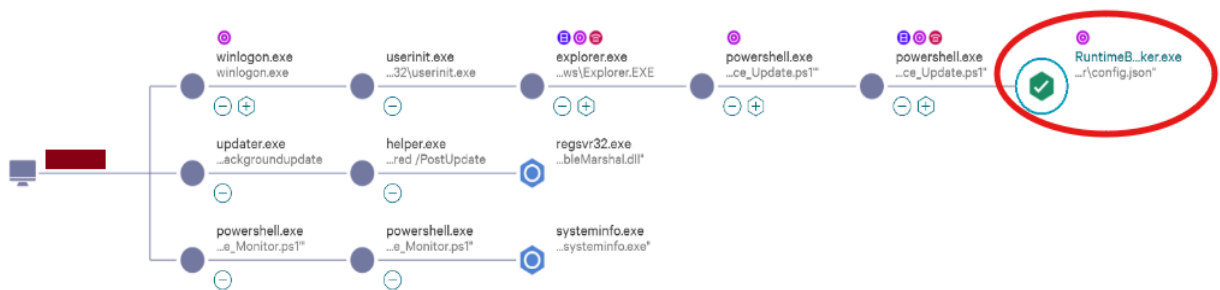
DETECTIONDETAILSRELATIONSCOMMUNITY 1.1 K

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Security vendors' analysis ⓘ

Lumu

Malware



Process: RuntimeBroker.exe

Start time: May 4, 2026 07:15:50

End time: May 4, 2026 07:15:50

Executable SHA256

f07c7223fdb691acbf0ebc7d9cc2ae614c0cf705920420c0130248a0c0e861d4

[Result here] <https://www.virustotal.com/gui/file/f07c7223fdb691acbf0ebc7d9cc2ae614c0cf705920420c0130248a0c0e861d4>

60

Community Score

60/70 security vendors flagged this file as malicious

f07c7223fdb691acbf0ebc7d9cc2ae614c0cf705920420c0130248a0c0e861d4

xmrig.exe

peexe idle 64bits

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY 13+

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Popular threat label

miner.bitminer/r002c0wn23

Threat categories

miner trojan pua

Security vendors' analysis

Acronis (Static ML)	Suspicious
Alibaba	Trojan:Win32/Coinminer.449
ALYac	Gen:Variant.Application.Miner.2
Arcabit	Trojan.Application.Miner.2

This process represents the final intended payload of the upper execution tree and was immediately terminated by security controls. The execution chain exhibits definitive indicators of T1036 (Defense Evasion via Masquerading). While **RuntimeBroker.exe** is a legitimate Windows core system binary, its execution out of an anomalous user profile subfolder (**\Device\HarddiskVolume2\Users\[redacted]\AppData\Local\Microsoft\WindowsApps\RuntimeBroker\RuntimeBroker.exe**) rather than the native trusted **C:\Windows\System32** directory confirms binary impersonation.



CrowdStrike flagged:

Detection: Machine Learning via Sensor-based ML

Detection Timestamp: May 4, 2026 07:16:52

Action taken: File quarantined + File blocked

Triggering indicator :

File path

`\Device\HarddiskVolume2\Users\[redacted]\AppData\Local\Microsoft\WindowsApps\RuntimeBroker\RuntimeBroker.exe`

The local on-sensor AI engine flagged the execution attempt due to real-time behavioral anomalies, triggering a pre-emptive termination based on localized heuristics.

```
PatternDispositionDescription: Block Process + Quarantine File + Response Action Already Applied
.ProcessBlocked: true
.QuarantineFile: true
```



CrowdStrike flagged:

Detection: Defense Evasion via Masquerading

Action taken: File quarantined + File blocked + Remediation performed

Associated MD5

`118c2d536d52dd30116baaf06dfe5e63`

The binary actively mimicked a core operating system component while operating from an unprivileged, non-standard path, establishing a clear structural signature of directory and binary masquerading.

```
PatternDispositionDescription: Prevention/Quarantine, process was blocked from execution and quarantine was attempted.
.ProcessBlocked: true
.QuarantineFile: true
```



Identified by CrowdStrike as an automated lead:

Detection: Machine Learning via Cloud-based ML

Detection Timestamp: May 4, 2026 14:59:34

Command line

```
"C:\Users\[redacted]\AppData\Local\Microsoft\WindowsApps\RuntimeBroker\RuntimeBroker.exe" -c
"C:\Users\[redacted]\AppData\Local\Microsoft\WindowsApps\RuntimeBroker\config.json"
```

File path

`\Device\HarddiskVolume2\Users\[redacted]\AppData\Local\Microsoft\WindowsApps\RuntimeBroker\RuntimeBroker.exe`

Cloud-based intelligence confirmed advanced evasion indicators, noting high mathematical entropy and binary packing anomalies characteristic of custom crypted payloads.

♦ Telemetry & SIEM Validation (Advanced Event Search):

Event Type: **AssociateTreeIdWithRoot**

Enforcement Timeline:

- **07:15:50.627** — Triggered **PatternId: 10139** (Behavioral Masquerading Context Correlation).
- **07:15:51.096** — Triggered **PatternId: 5700** (Sensor-Based ML Real-Time Execution Halt).

May. 4, 2026 7:15:50.627

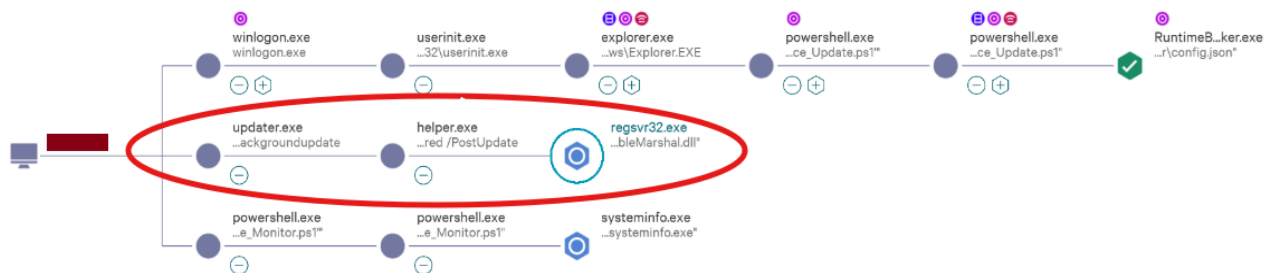
"PatternId": "10139"

May. 4, 2026 7:15:51.096

"PatternId": "5700"

Observed Command 2:

Ruled Out Context (False Positive)



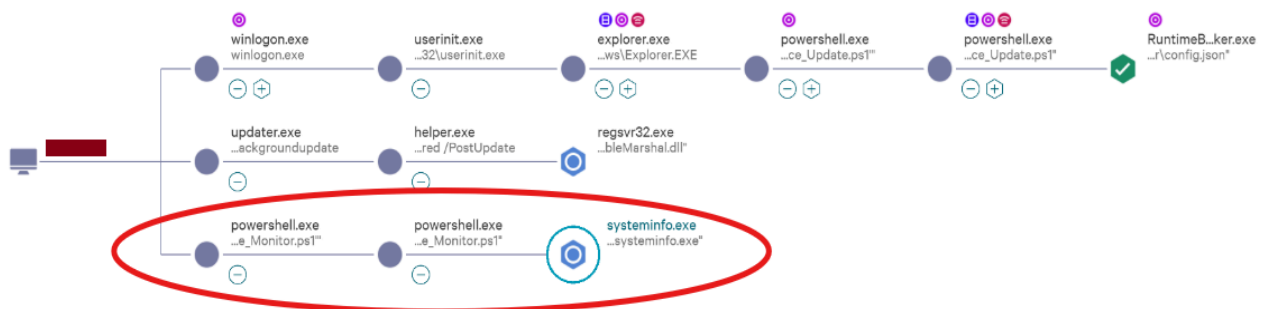
Processes: **updat.exe** → **helper.exe** → **regsvr32.exe**

Command Line: "C:\Windows\system32\regsvr32.exe" /s
"C:\Users\[redacted]\AppData\Local\Mozilla
Firefox\AccessibleMarshal.dll"

This alert is classified as a False Positive. The **regsvr32.exe** utility falls under the LOLBAS (Living off the Land Binaries and Scripts) classification and is heavily abused by threat actors to execute arbitrary code silently. Consequently, its invocation within a browser-related process subtree automatically triggers standard heuristic telemetry rules within CrowdStrike Falcon.

However, deep context verification validates its benign nature: the system registration binary was executed in silent mode (/s) exclusively to register the official, digitally signed library **AccessibleMarshal.dll** (a legitimate Mozilla component responsible for Accessibility features). The zero-enforcement state (Actions taken: None) and the complete absence of this event within the telemetry detection repository (#repo:detections) confirm that this process posed no security threat, had no correlation with the malicious XMRig deployment, and constitutes standard operating system background noise.

Observed Command 3:



② Phase 2 : Host Reconnaissance & Discovery

Process: **powershell.exe** (1 - Discovery Policy Evasion)

Start time: May 5, 2026 12:01:29

End time: May 5, 2026 12:01:34

Command line

```
"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"  
"-Command" "if((Get-ExecutionPolicy) -ne 'AllSigned') {  
Set-ExecutionPolicy -Scope Process Bypass }; &  
'C:\Users\[redacted]\Downloads\System_Performance_Monitor.ps1'"
```

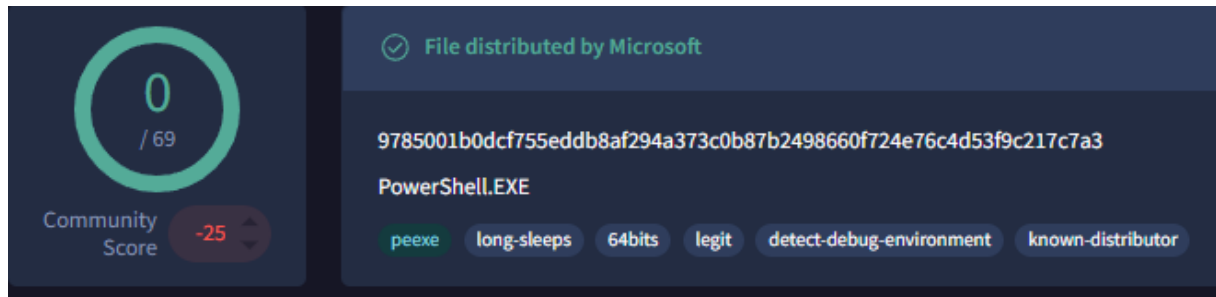
File path

```
\Device\HarddiskVolume2\Windows\System32\WindowsPowerShell\v1.  
0\powershell.exe
```

Executable SHA256

9785001b0dcf755eddb8af294a373c0b87b2498660f724e76c4d53f9c217c7a3

[Result there] <https://www.virustotal.com/gui/file/9785001b0dcf755eddb8af294a373c0b87b2498660f724e76c4d53f9c217c7a3>



Mirroring the defense evasion techniques seen in Phase 1, this transient 5-second instance was spawned to systematically check and dismantle execution policy restrictions. By forcing an inline **-Scope Process Bypass**, the adversary primed the environment to execute the untrusted script **System_Performance_Monitor.ps1** from the user's Downloads folder without changing permanent registry configurations.

Process: **powershell.exe** (2 - Reconnaissance Script Engine)

Start time: May 5, 2026 12:01:34

End time: May 5, 2026 12:02:25

Command line

```
"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"  
-NoProfile -ExecutionPolicy Bypass -File  
"C:\Users\[redacted]\Downloads\System_Performance_Monitor.ps1"
```

File path

```
\Device\HarddiskVolume2\Windows\System32\WindowsPowerShell\v1.  
0\powershell.exe
```

This process acted as the core execution vehicle for the second stage script. Running with **-NoProfile** to minimize logging footprints, the script remained active in memory for 51 seconds. The name **System_Performance_Monitor.ps1** represents a deceptive masquerading attempt designed to mimic normal system administrative workflows. The primary objective of this script session was to systematically spawn discovery commands to audit the host machine configuration.

Process: systeminfo.exe

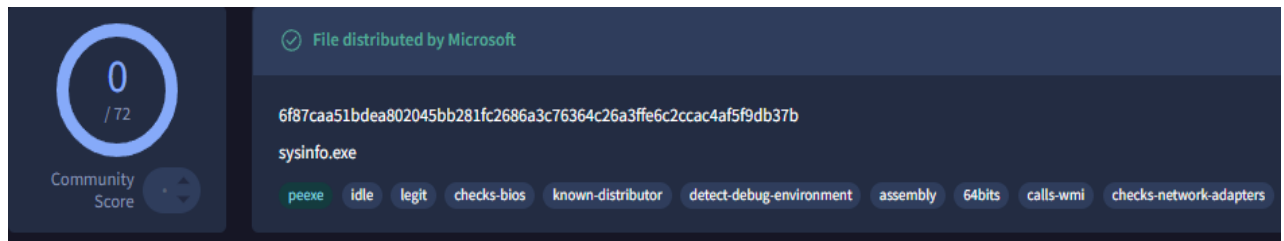
Start time: May 5, 2026 12:01:47

End time: May 5, 2026 12:01:53

Executable SHA256

6f87caa51bdea802045bb281fc2686a3c76364c26a3ffe6c2ccac4af5f9db37b

[Result there|<https://www.virustotal.com/gui/file/6f87caa51bdea802045bb281fc2686a3c76364c26a3ffe6c2ccac4af5f9db37b>]



Identified by CrowdStrike as an automated lead:

Detection: Discovery via System Information Discovery

Detection Timestamp: May 5, 2026 12:02:51

Command line

"C:\Windows\system32\systeminfo.exe"

File path

\Device\HarddiskVolume2\Windows\System32\systeminfo.exe

This activity constitutes confirmed Host Reconnaissance. While **systeminfo.exe** is a legitimate, native Windows administrative binary (0/75 on VirusTotal), its execution from the context of an unvetted user download script confirms malicious intent. The adversary explicitly abused this tool to collect critical system parameters, including OS architecture, domain naming structures, hardware specifications, network interfaces, and installed hotfixes/patches.

Threat actors routinely leverage this precise information to identify missing security updates, customize subsequent privilege escalation exploits, or determine if the machine is a sandboxed analysis environment. Though CrowdStrike registered this as a non-enforced context lead (**Actions taken: None**), it confirms that the adversary successfully collected architectural layout data from the [redacted-host] endpoint before this specific tool session closed.

Analysis

A thorough analysis of associated processes (lsass, services, svchost) has been conducted. No anomalous activity indicating additional attack vectors or hidden backdoors within system processes was detected. All network and registry activity within these branches has been deemed legitimate system background activity (baseline).

Command Line Breakdown: Phase 1 (Initial Access & Payload Deployment)

"C:\Windows\System32\cmd.exe" /q /c del /q "C:\Program Files\Microsoft OneDrive\Update\OneDriveSetup.exe"

- **Indicator Removal** → The adversary subverted the native Windows **RunOnce** registry key via **runonce.exe** to quietly execute targeted file deletions. The use of silent flags (**/q /c del**) was optimized to purge the source installer artifacts and corrupt the forensic timeline.

"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -w hidden -ep bypass -File "C:\Users\[redacted]\AppData\Local\Temp\firefox_dump.zip"

- **-w hidden & -ep bypass** → Standard adversary tradecraft to completely hide the desktop console window from the active user and force an execution policy override.
- **Defense Evasion** → The malicious loader script was masqueraded inside a fake compressed archive (**firefox_dump.zip**) and executed directly from the user's local **Temp** path to evade basic static string inspection.

"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -NoProfile -ExecutionPolicy Bypass -File "C:\Users\[redacted]\Downloads\Runtime_Service_Update.ps1"

- **Resource Hijacking** → The core script engine established automated external connections to GitHub infrastructure to ingest a cryptocurrency mining suite (**xmrig.exe**) and drop a low-level hardware access kernel driver (**WinRing0x64.sys**) into user-space directories.
- **Evasive Persistence** → Simultaneously, the script initiated compilation sequences to compile-write an untrusted backdoor payload disguised as a core Windows component directly into an anomalous local application path.

"C:\Users\[redacted]\AppData\Local\Microsoft\WindowsApps\RuntimeBroker\RuntimeBroker.exe" -c

"C:\Users\[redacted]\AppData\Local\Microsoft\WindowsApps\RuntimeBroker\config.json"

- **Process Masquerading** → The binary explicitly mimicked the name of a legitimate Microsoft core system component (**RuntimeBroker.exe**) to blend with benign background processes while loading a hidden configuration layer (**-c config.json**).
- **Software Packing/Entropy** → Cloud-based intelligence flagged high mathematical entropy and binary packing configurations, indicating a heavily obfuscated defense evasion attempt.

Command Line Breakdown: Phase 2 (Host Reconnaissance & Discovery)

"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -NoProfile -ExecutionPolicy Bypass -File "C:\Users\[redacted]\Downloads\System_Performance_Monitor.ps1"

- **Execution Pattern Continuity** → The adversary reappeared approximately 16 hours later, utilizing an identical process-scope execution policy bypass rule. The script was intentionally renamed to **System_Performance_Monitor.ps1** to masquerade as legitimate administrative workflows and lower the risk of detection during active runtime monitoring.

"C:\Windows\system32\systeminfo.exe"

- **System Information Discovery** → The script abused the native Microsoft configuration auditing tool to harvest detailed host parameters, including OS architecture, active domain trees, hardware specifications, and hotfix patch levels. This reconnaissance data is routinely gathered to map vulnerabilities for subsequent privilege escalation exploits.

The Verdict (SUCCESSFUL BLOCK) → Raw telemetry confirms a synchronized multi-layered enforcement action. Local hybrid engines successfully intercepted the execution within milliseconds (**PatternId: 5700** and **10139**). The masqueraded **RuntimeBroker.exe** binary was completely blocked and shifted into a quarantined state before it could establish persistent external C2 communications.

Potential Impact

1. **Unauthorised Resource Hijacking (Partial Success):** The core PowerShell script successfully dropped a cryptocurrency mining suite (xmrig.exe, solo_mine_example.cmd) and a low-level hardware kernel driver (WinRing0x64.sys) into the local user Temp directory. While the primary backdoor was intercepted, these staging files remain on disk and pose an active risk to system performance and resource integrity until manually purged.
2. **Host Infrastructure Reconnaissance (Success):** The execution of systeminfo.exe via the second-stage script (System_Performance_Monitor.ps1) was successfully completed with no enforcement action taken (Actions taken: None). The adversary successfully gathered the workstation's OS architecture, domain naming schema, and hotfix patch levels, effectively mapping out the system baseline to identify potential privilege escalation exploits.
3. **Forensic Artifact Tampering (Success):** The adversary successfully subverted the Windows Auto-Start Execution Points (ASEP) by leveraging the RunOnce registry key to trigger runonce.exe and cmd.exe. This allowed them to quietly purge the source OneDriveSetup.exe installers from the program directories, intentionally eliminating early-stage indicators of compromise and disrupting forensic timeline reconstruction.

4. **Evasive Backdoor Persistence (Prevented):** The compilation and attempted deployment of the packed RuntimeBroker.exe implant within the WindowsApps folder was completely countered. Falcon's hybrid machine learning and behavioral masquerading engines intercepted the execution within milliseconds, preventing the attacker from establishing a persistent, hidden command-and-control (C2) channel via the config.json layer.
 5. **Threat Containment Velocity (Near-Zero Payload Dwell Time):** Due to automated EDR intervention at the critical execution threshold of the masqueraded payload, the adversary failed to transition from script execution to an interactive binary shell. The dwell time for the primary backdoor implant was effectively restricted to less than one second, preventing any actual damage to corporate data integrity.
-

Host Inspection:

During the manual investigation of host [redacted-host] , the following artifacts and evidence were identified, confirming the intrusion and execution chain.

1. **Filesystem Artifacts (Evidence of Staging)** A manual search of the user's directory confirmed the presence of the malicious directory structure and persistent configuration payload identified in the CrowdStrike alerts.

Directory: **C:\Users\[redacted]\AppData\Local\Microsoft\WindowsApps\RuntimeBroker**
File Identified: **config.json**

SHA256: 37BE1EB4641F5A6061648623B289BFFCAD863BE47F5031F6980B542A09904EB1

VirusTotal Score: 0/75 (Unknown/Unseen). Classified as a newly generated, campaign-specific mining profile.

2. **Configuration Content Analysis (Static Analysis)** Manual inspection of the configuration logic (via cat) revealed the adversary's automated deployment process and mining pool destination parameters: Payload Intent: The parameters explicitly map out an unauthorized XMRig cryptocurrency mining operational environment. Resource Exploitation: The configuration was optimized to heavily hijack workstation performance via the **"max-threads-hint": 80** key, forcing up to 80% CPU utilization. C2 Pool Reporting: Connections were mapped to an external stratum mining pool destination at **portugalloindostan.top:443** using strict TLS encryption (**"tls": true**) to bypass network detection controls, funneling assets directly to a hardcoded attacker wallet address.

```
C:\Users\████████\AppData\Local\Microsoft\WindowsApps\RuntimeBroker> cat config.json

{
  "autosave": true,
  "cpu": {
    "enabled": true,
    "max-threads-hint": 80
  },
  "pools": [
    {
      "url": "portugalloindostan.top:443",
      "user": "48edfHu7V9Z84YzzMa6fUueoELZ9ZRXq9VetWzYGzKt52XU5xvqgzYnDK9URnRoJMk1j8nLwEVsaSWJ4fhdUyZijBGUicoD",
      "pass": "x",
      "tls": true
    }
  ]
}
```

Execution Evidence (Command Line Telemetry) Forensic evidence from the host's memory and event logs provided the exact parameters used to initiate the mining impact: Mining Command: **"C:\Users\[redacted]\AppData\Local\Microsoft\WindowsApps\RuntimeBroker\RuntimeBroker.exe" -c "C:\Users\[redacted]\AppData\Local\Microsoft\WindowsApps\RuntimeBroker\config.json"**

Analysis: The binary deliberately targeted system binary impersonation to run out of an unprivileged user folder. The use of the **-c** switch to load the local configuration file confirms that the masqueraded process execution and the cryptomining configuration block were part of the same coordinated attack.

System Remediation State A check of the temporary directories and the EDR quarantine status confirmed: Quarantined Binaries: RuntimeBroker.exe (confirmed successfully extracted and isolated from the **WindowsApps\RuntimeBroker** path).

Neutralized Drivers: WinRing0x64.sys, xmrig.exe, and solo_mine_example.cmd (staged within the local user **Temp** path). Current State: No active malicious processes are currently running on the host. The primary backdoor execution tree was successfully defeated by the EDR sensor. Remaining passive folder structures and the **config.json** text artifact are queued for final manual purging.

[Resultthere|<https://www.virustotal.com/gui/search?query=37BE1EB4641F5A6061648623B289BFFCAD863BE47F5031F6980B542A09904EB1>]

```
C:\> cd C:\Users\████████\AppData\Local\Microsoft\WindowsApps\RuntimeBroker\

C:\Users\████████\AppData\Local\Microsoft\WindowsApps\RuntimeBroker
C:\Users\████████\AppData\Local\Microsoft\WindowsApps\RuntimeBroker> ls

Directory listing for C:\Users\████████\AppData\Local\Microsoft\WindowsApps\RuntimeBroker -

Name                               Type                Size (bytes)        Size (MB) Last Modified (UTC-4)    Created (UTC-4)
----                               -
config.json                        .json               361                 0 5/4/2026 7:15:50 AM    5/4/2026 7:15:50 AM

C:\Users\████████\AppData\Local\Microsoft\WindowsApps\RuntimeBroker> filehash C:\Users\finance\AppData\Local\Microsoft\WindowsApps\RuntimeBroker\config.json

Filename : C:\Users\████████\AppData\Local\Microsoft\WindowsApps\RuntimeBroker\config.json
MD5      : 848FD635E437068EF4E2C7CAA8688DF2
SHA1     : 8082512190015214231E9007E8E4829AA6A208DE
SHA256   : 37BE1EB4641F5A6061648623B289BFFCAD863BE47F5031F6980B542A09904EB1
```

Response:

CrowdStrike Falcon detection page for the **RuntimeBroker.exe** event — Action taken: **Process blocked + File quarantined** (Purged). The primary masqueraded backdoor payload was neutralized (SHA256: **f07c7223fdb691acb0ebc7d9cc2ae614c0cf705920420c0130248a0c0e861d4**). As documented in 1 screenshot and 2, the binary was delivered via an evasive PowerShell stager and triggered both the behavioral engine for File Impersonation (T1036) and the Machine Learning sensor (CST0007).

Process · [See more details](#)

Process	Actions taken				
RuntimeBroker.exe	Process blocked				
	File quarantined				
Severity	Objective	Tactic	Technique	Technique ID	
Low	Falcon Detection Method	Machine Learning	Sensor-based ML	CST0007	
Description					
This file meets the machine learning-based on-sensor AV protection's low confidence threshold for malicious files.					
Command line					
"C:\Users\████████\AppData\Local\Microsoft\WindowsApps\RuntimeBroker\RuntimeBroker.exe" -c "C:\Users\████████\AppData\Local\Microsoft\WindowsApps\RuntimeBroker\config.json"					

File path

[\Device\HarddiskVolume2\Users\████████\AppData\Local\Microsoft\WindowsApps\RuntimeBroker\RuntimeBroker.exe](#)

Process · [See more details](#)

Process	Actions taken				
RuntimeBroker.exe	Process blocked				
	File quarantined				
	Remediation performed 🔗				
Severity	Objective	Tactic	Technique	Technique ID	
Medium	Keep Access			T1036	
Description					
A file appears to be imitating a standard OS or otherwise benign filename and/or launched from an unusual location. This might be to masquerade malware. Review the file.					
Command line					
"C:\Users\████████\AppData\Local\Microsoft\WindowsApps\RuntimeBroker\RuntimeBroker.exe" -c "C:\Users\████████\AppData\Local\Microsoft\WindowsApps\RuntimeBroker\config.json"					

File path

[\Device\HarddiskVolume2\Users\████████\AppData\Local\Microsoft\WindowsApps\RuntimeBroker\RuntimeBroker.exe](#)

Execution was terminated and the binary status was updated to Purged at 07:15:53, as confirmed by Screenshot 3.



Quarantined files

Hash

f07c7223fdb691acbf0ebc7d9cc2ae614c0cf705920420c0130248a0c0e861d4

Time quarantined

May 4, 2026 07:15:53

Status

Purged

Filename

RuntimeBroker.exe

[See more in Quarantined files](#)

Automated sandbox analysis detailed in Screenshot 4.



Sandbox analysis

Submitted by

Falcon Intelligence (automatically)

Detonated

May 4, 2026 07:16:56

Threat score

100/100

Verdict

malicious

Description

PE32+ executable for MS Windows 6.00 (console), x86-64, 10 sections

Evasive

The input sample contains a known anti-VM trick

[See more in Sandbox reports](#)

Screenshot 5, and Screenshot 6 gave the file a Malicious Threat Score of 100/100, identifying embedded anti-VM tricks and explicitly classifying the payload with the **"Miner"** tag.

Sandbox Report

File: f07c7223fdb691acbf0ebc7d9cc2ae614c0cf705920420c0130248a0c0e861d4



Malicious

100/100

Threat details

Threat level



Malicious

Threat score

100/100

Description

f07c7223fdb691acbf0ebc7d9cc2ae614c0cf705920420c0130248a0c0e861d4 has a malicious verdict as it had a threat score between 50 and 100. This is due to a high amount of matching signatures during analysis, of which some have properties such as having a high relevance or being a monitored process that increases their contribution to the threat score.

Also, f07c7223fdb691acbf0ebc7d9cc2ae614c0cf705920420c0130248a0c0e861d4 may have a high similarity with other malicious samples observed, or a direct existing sample match within our repository.

Findings

Behavior threat indicators

- Malicious (2)
- Suspicious (17)
- Informative (155)

Discovered

Tags

Miner

Unlike the compiled binary, the unprivileged configuration script `config.json`, the initial staging archive `firefox_dump.zip` in the local Temp path, and the second-stage reconnaissance script `System_Performance_Monitor.ps1` in the user's Downloads directory were not automatically removed and remain on the host. Manual remediation is required to delete the residual unprivileged application directory `C:\Users\[redacted]\AppData\Local\Microsoft\WindowsApps\RuntimeBroker`, purge the dormant XMRig staging files (`xmrig.exe`, `solo_mine_example.cmd`, `WinRing0x64.sys`) from the Temp folder, and clean the user download paths to eliminate remaining indicators of compromise. Additionally, a network firewall block and outbound log audit for the stratum mining pool domain `portugalloindostan.top` over port 443 are recommended due to the hardcoded C2 infrastructure extracted from the surviving configuration block.

Overall Analyst Assessment

This incident is assessed as a True Positive (TP) involving a multi-stage, targeted resource hijacking and reconnaissance campaign targeting the [redacted-host] workstation. The adversary gained initial access via credential compromise and RDP session hijacking, reconnecting to a long-standing, persistent REMOTE_INTERACTIVE session belonging to the [redacted] user at 06:27:19. This allowed the attacker to completely bypass primary authentication gateways and EDR login triggers.

To ensure early defense evasion and disrupt forensic tracking, the attacker immediately subverted Windows Auto-Start Execution Points (ASEP) by hijacking the RunOnce registry key to spawn `runonce.exe` and `cmd.exe` instances, successfully purging the source `OneDriveSetup.exe` installer files before advanced baseline monitors could alert on the source. At 06:29, the adversary executed a hidden PowerShell stager disguised as a compressed archive (`firefox_dump.zip`) within the user Temp directory to harvest browser-stored credentials.

To facilitate secondary payload delivery within the hijacked desktop session, the adversary launched the local mail client (`thunderbird.exe`) at 07:13, forcing an encrypted IMAP synchronization to ingest a targeted malicious shortcut payload (`Runtime_Service_Update.ps1.lnk`) into the local profile database. The execution of this shortcut triggered an automated retrieval script from GitHub to stage the core XMRig mining suite (`xmrig.exe`, `solo_mine_example.cmd`, and the low-level `WinRing0x64.sys` driver) into the local user space.

Simultaneously, the script performed advanced directory and binary Masquerading (T1036) by compile-writing a packed backdoor implant disguised as a core system component (`RuntimeBroker.exe`) into an anomalous application folder. CrowdStrike Falcon successfully intercepted the execution via its behavioral masquerading engine (PatternId 10139) and localized machine learning (PatternId 5700), blocking the process in less than one second and completely purging the malicious binary (SHA256: `f07c7223fdb691acbf0ebc7d9cc2ae614c0cf705920420c0130248a0c0e861d4`). Automated sandbox testing validated a maximum malicious score of 100/100, identifying active anti-VM evasion mechanisms.

Advanced host and telemetry triage confirmed that approximately 16 hours following the initial block, the adversary re-engaged the host using an identical process-scope execution policy bypass pattern. Operating under a deceptive administrative file name (`System_Performance_Monitor.ps1`) from the user's Downloads folder, the attacker

successfully executed `systeminfo.exe` (Actions taken: None) to map the system's architecture, domain layout, and hotfix patch baseline to identify privilege escalation paths.

While the interactive backdoor threat was completely neutralized, the incident is only partially contained. Passive files including the custom `config.json` layer—which contains a hardcoded, encrypted stratum mining pool destination (`portugalloindostan.top:443`) and the actor's verified Monero wallet address (`48edfHu7V9Z84YzzMa6fUueoELZ9ZRXq9VetWzYGzKt52XU5xvqgzYnDK9URNroJMk1j8nLwEVsaSWJ4fhdUyZijBGUicoD`)—remain on disk alongside the staging PowerShell files. Immediate network isolation and manual remediation are required to purge these residual filesystem artifacts and implement perimeter blocks against the extracted network infrastructure.

Mini Attack Visualization

[Phase 1: Initial Access & Payload Deployment]

↓

[Interactive Session Active: user '`[redacted]`' on host `[redacted-host]`]

↓

[Defense Evasion: `runonce.exe` & `cmd.exe` execute to purge early OneDrive setup artifacts]

↓

[Execution: Hidden `powershell.exe` instance invokes stager disguised as `firefox_dump.zip`]

↓

[Post-Compromise Tradecraft: `thunderbird.exe` processes pre-staged payload via encrypted IMAP sync]

↓

[Staging: `Runtime_Service_Update.ps1` bypasses execution policy & streams XMRig miner from GitHub]

↓

[Masquerading: Script drops a packed, custom backdoor implant named `RuntimeBroker.exe` into `WindowsApps`]

↓

[Execution Attempt: `RuntimeBroker.exe` launches with `-c config.json` to hijack 80% CPU resources]

↓

[EDR Enforcement: Falcon Behavioral (PatternId 10139) & ML (PatternId 5700) engines block process and quarantine file (Purged)]

↓

[Phase 2 (~16hrs Later): Adversary re-engages host via disguised System_Performance_Monitor.ps1 script]

↓

[Host Discovery (Success): systeminfo.exe runs with zero enforcement to harvest OS patch & architecture baselines]

↓

[Current State: Partially Contained (RuntimeBroker.exe neutralized; passive config.json and staging scripts remain)]

Recommended Actions:

1. Immediate Host Containment & Isolation (CrowdStrike Falcon)

- **Network Isolation:** Trigger the automated network containment feature via the CrowdStrike Falcon console for host [redacted-host] immediately. This blocks all inbound and outbound network traffic (preventing Lateral Movement and automated downloading of secondary payloads) while preserving the EDR telemetry channel for investigation.
- **Registry & Persistence Freeze:** Maintain containment until all subsequent credential revocations and volatile memory sweeps are completed.

2. Access & Credential Security

- **Password Reset:** Enforce an immediate, mandatory complex password change for the [redacted-domain] domain account. Since a hidden PowerShell stager named **firefox_dump.zip** was executed within the user profile context at 06:29, assume all browser-stored credentials and session cookies are compromised.
- **Session Termination:** Revoke and terminate all active interactive, RDP, and mail application sessions for the [redacted] user across the network to invalidate any exfiltrated session tokens and kick the adversary out of the REMOTE_INTERACTIVE session.
- **RDP Hardening:** Restrict RDP access to the [redacted-host] workstation using Network Level Authentication (NLA), limit access to specific authorized IP ranges (Admin Jump-host), and implement MFA.

3. Filesystem Remediation (Post-Isolation Cleanup)

- **FileSystem Purging:** Once the host is network-isolated, delete the residual unprivileged application directory **C:\Users\[redacted]\AppData\Local\Microsoft\WindowsApps\RuntimeBroker** along with the surviving **config.json** file.

- **Staging Artifact Cleanup:** Scour and delete the dormant cryptocurrency mining files (**xmrig.exe**, **solo_mine_example.cmd**, **WinRing0x64.sys**) from the local user Temp path **C:\Users\[redacted]\AppData\Local\Temp**.
- **Script Deletion:** Locate and permanently remove the malicious script loader **Runtime_Service_Update.ps1** and the discovery script **System_Performance_Monitor.ps1** from the user's Downloads folder.
- **Indicator Audit:** Review the Windows Registry for unauthorized modifications within the Run and RunOnce keys (specifically inspecting entries triggered around 06:27 on May 4, 2026).

4. Network & IOC Controls

- **Infrastructure Blocking:** Blacklist the following malicious stratum mining pool destination and external ingestion endpoints at the perimeter firewall:
 - **portugalloindostan.top** (Malicious Cryptomining C2 Pool over Port 443)
 - **140.82.121.4 / 185.199.110.133** (External staging locations associated with payload distribution)
- **Fleet Hunting:** Perform an enterprise-wide endpoint sweep for the following high-fidelity Indicators of Compromise (IoCs):
 - **SHA256:**
f07c7223fdb691acbf0ebc7d9cc2ae614c0cf705920420c0130248a0c0e861d4 (Masqueraded RuntimeBroker.exe)
 - **SHA256:**
37be1eb4641f5a6061648623b289bffcadd863be47f5031f6980b542a09904eb1 (Malicious config.json profile)

5. System Hardening & GPO

- **PowerShell Policy Restriction:** Enforce a strict AllSigned script execution policy via Group Policy Objects (GPO) across the domain. Enable PowerShell Script Block Logging (Event ID 4104) to enhance deep-dive inspection visibility into obfuscated, process-scope execution bypass attempts.
- **User Space Hardening:** Implement explicit software restriction policies (such as AppLocker) to strictly block the execution of binary files or scripts launched directly from unprivileged locations (**AppData\Local** and **Downloads**).

6. Escalation & Deep Dive Analysis

- **RDP Access & Credential Root Cause Analysis (RCA):** Conduct an immediate forensic audit of the Domain Controller security logs and the local workstation's **Security.evtx** log. Target Event ID 4624 (specifically Logon Type 10 - RemoteInteractive or Type 3 - Network) matching the timestamp 06:27:19 on May 4, 2026. Extract the source IP address, workstation name, and routing path to identify the attacker's true network origin and determine if the credentials were leaked via an external breach, info-stealer infection, or compromised via brute-force (Event ID 4625).
- **EDR Policy Review:** Coordinate with the Security Engineering team to evaluate why the execution of **systeminfo.exe** during Phase 2 discovery did not trigger automated containment or alert generation on the sensor earlier in the triage timeline.

- **Threat Hunting Handoff:** Provide the unique cryptocurrency wallet address extracted from the configuration block
(48edfHu7V9Z84YzzMa6fUueoELZ9ZRXq9VetWzYGzKt52XU5xvqgzYnDK9URNroJMk1j8nLwEVsaSWJ4fhdUyZijBGUicoD) to the Cyber Threat Intelligence (CTI) team to monitor for broader campaigns targeting corporate resources.